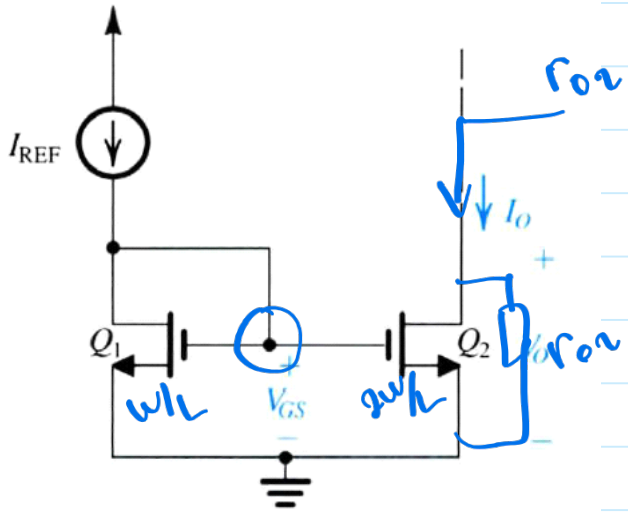
Reading

Adel S. Sedra and Kenneth C. (KC) Smith, Tony Chan Carusone, and Vincent Gaudet
Microelectronic Circuits 8th Edition, *Oxford University Press*, 2019.

- Chapter 8.5 The Cascode Amplifier
- Chapter 8.6 The IC Source Follower
- Chapter 8.7 Current Mirror Circuits with Improved Performance

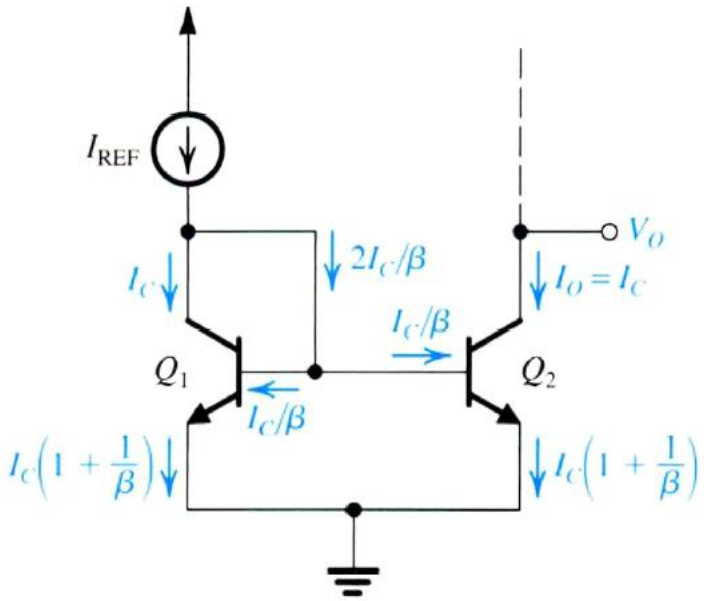
Recap



$$I_O|_{\text{nominal}} = \frac{(W/L)_2}{(W/L)_1} I_{REF}$$

$$I_O = \frac{(W/L)_2}{(W/L)_1} I_{REF}\left[1 + \frac{V_O - V_{GS}}{V_{A2}}\right]$$

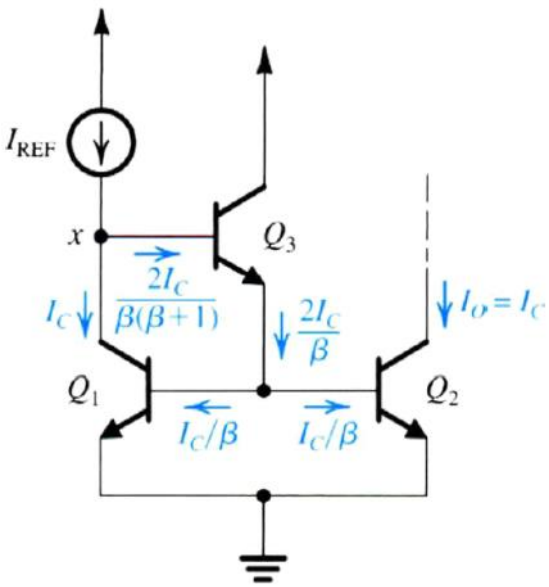
$$R_o = r_{o2}$$



$$\frac{I_O}{I_{REF}} = \frac{I_C}{I_C\left(1+\frac{2}{\beta}\right)} = \frac{1}{1+\frac{2}{\beta}}$$

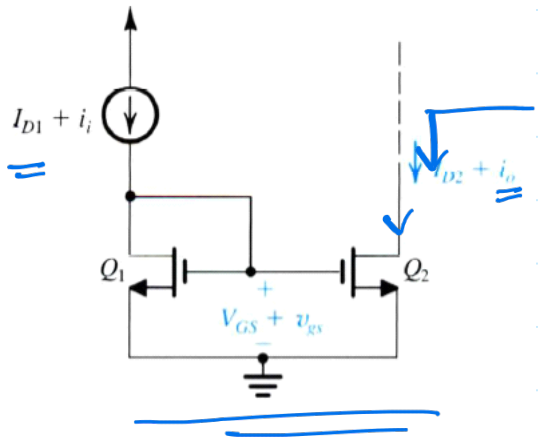
$$\frac{I_O}{I_{REF}} = \frac{m}{1+\frac{m+1}{\beta}}$$

$$I_O = I_{REF}\frac{m}{1+\frac{m+1}{\beta}}\left(1+\frac{V_O - V_{BE}}{V_{A2}}\right)$$

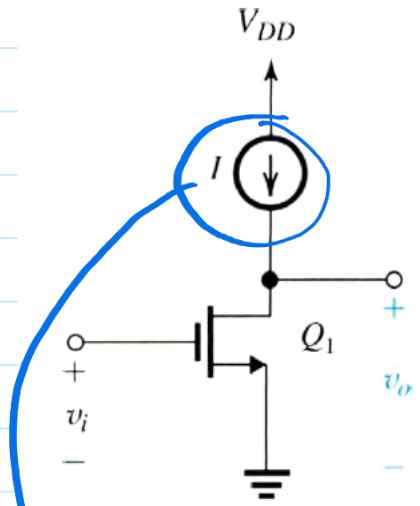


$$\frac{I_O}{I_{REF}} = \frac{1}{1+2/(\beta^2+\beta)} \simeq \frac{1}{1+2/\beta^2}$$

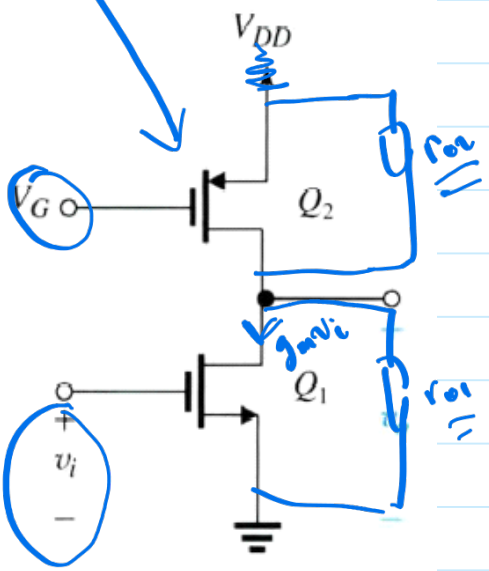
Recap



$$R_o = r_{o2}$$



$$A_{vo} = -g_m r_o$$

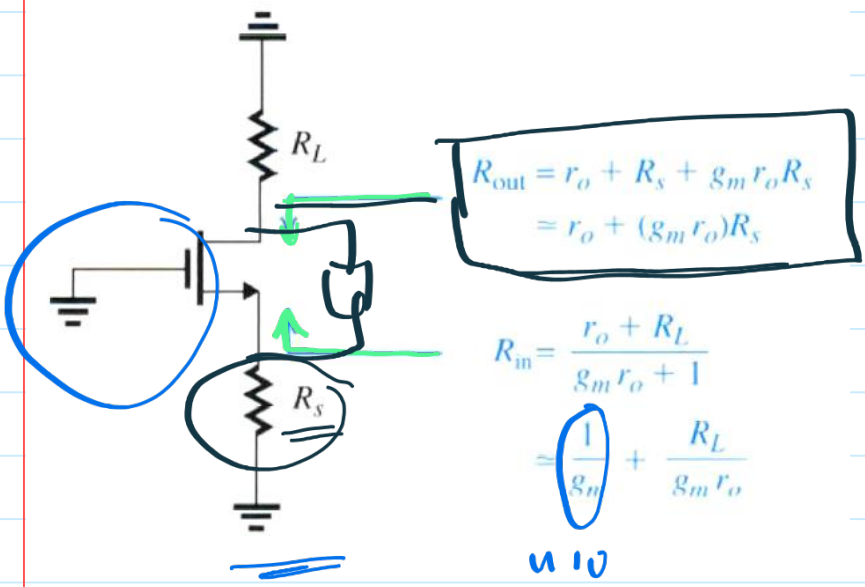


$$A_v \equiv \frac{v_o}{v_i} = -g_{m1}(r_{o1} \| r_{o2})$$

$$A_v = -\frac{1}{2} g_m r_o$$

Recap

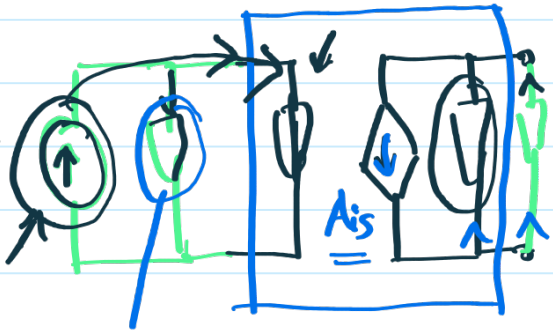
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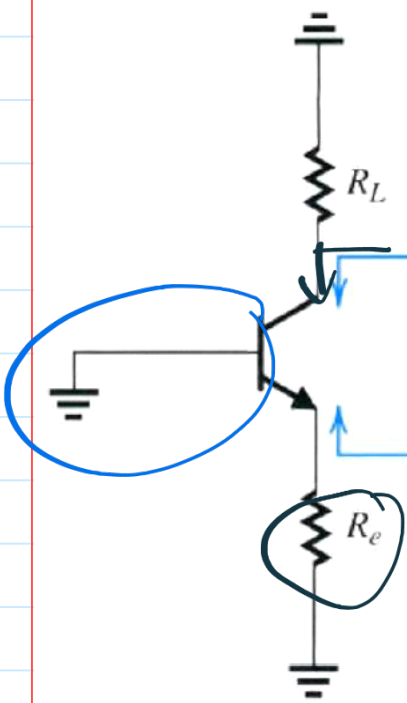


$$R_{out} = r_o + R_s + g_m r_o R_s$$
$$= r_o + (g_m r_o) R_s$$

$$R_{in} = \frac{r_o + R_L}{g_m r_o + 1}$$
$$\simeq \frac{1}{g_m} + \frac{R_L}{g_m r_o}$$

u 1 0



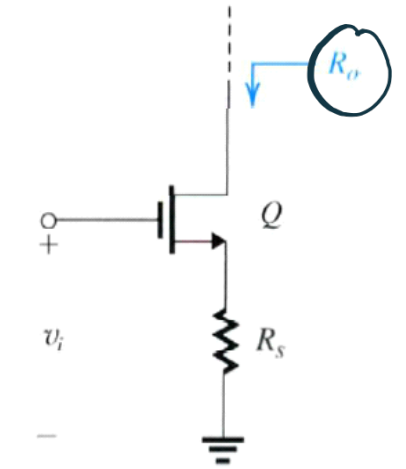


$$R_{out} = r_o + (R_e \| r_\pi) + g_m r_o (R_e \| r_\pi)$$
$$\simeq r_o + (g_m r_o)(R_e \| r_\pi)$$

$$R_{in} = r_\pi \| \left[\frac{r_o + R_L}{g_m r_o + 1} \right]$$
$$\simeq \frac{1}{g_m} + \frac{R_L}{g_m r_o}$$

C B

Output Resistance of CS/CE with Rs/Re



$$R_o = R_s + r_o + g_m r_o R_s$$

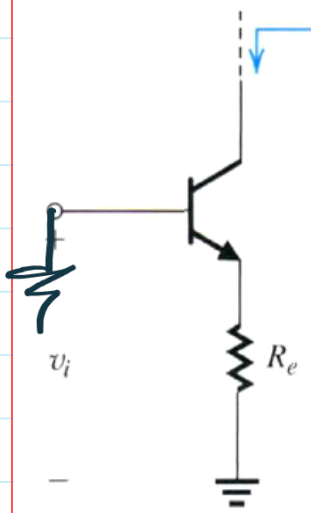
$$R_o \simeq (1 + g_m R_s)\, r_o$$

CS w Rs



$$R_o = r_o + (R_e \| r_\pi) + g_m r_o (R_e \| r_\pi)$$

$$\simeq r_o[1 + g_m (R_e \| r_\pi)]$$



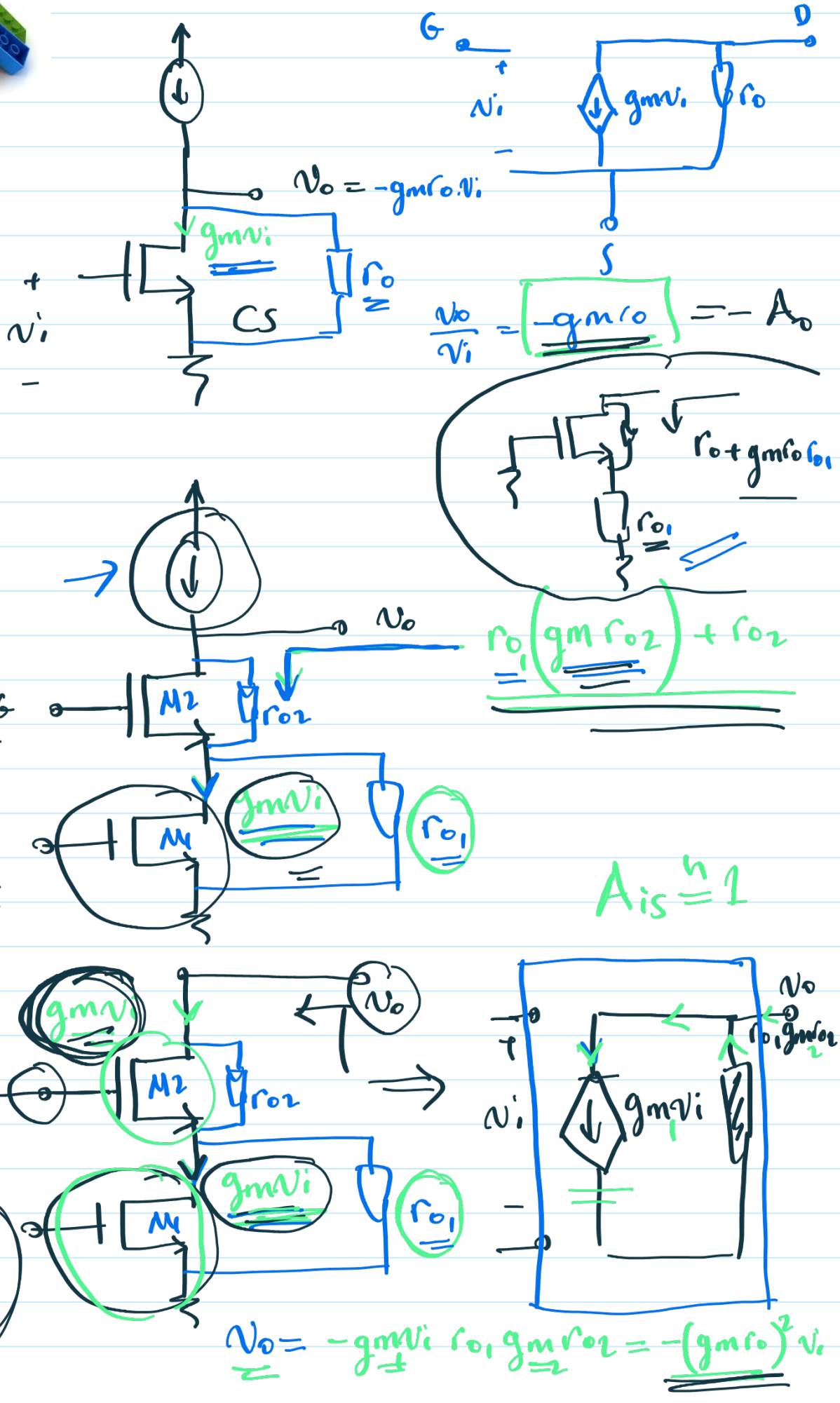
CE w Re

=

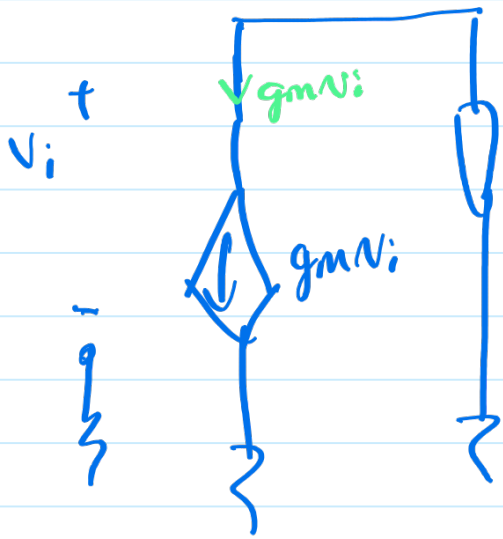
CS w Rs

The Cascode Amplifier

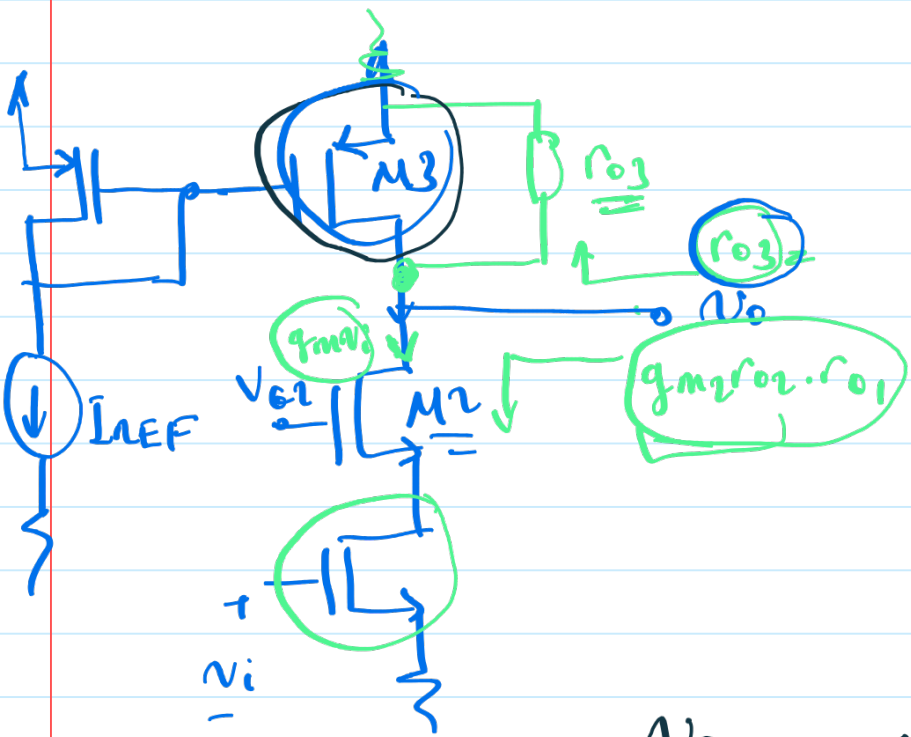


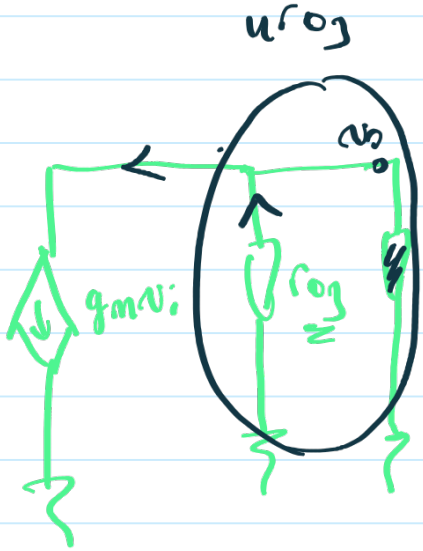


The Cascode Amplifier with CS Load



$$A_v = -g_{m1} r_{o1} g_{m2} r_{o2}$$

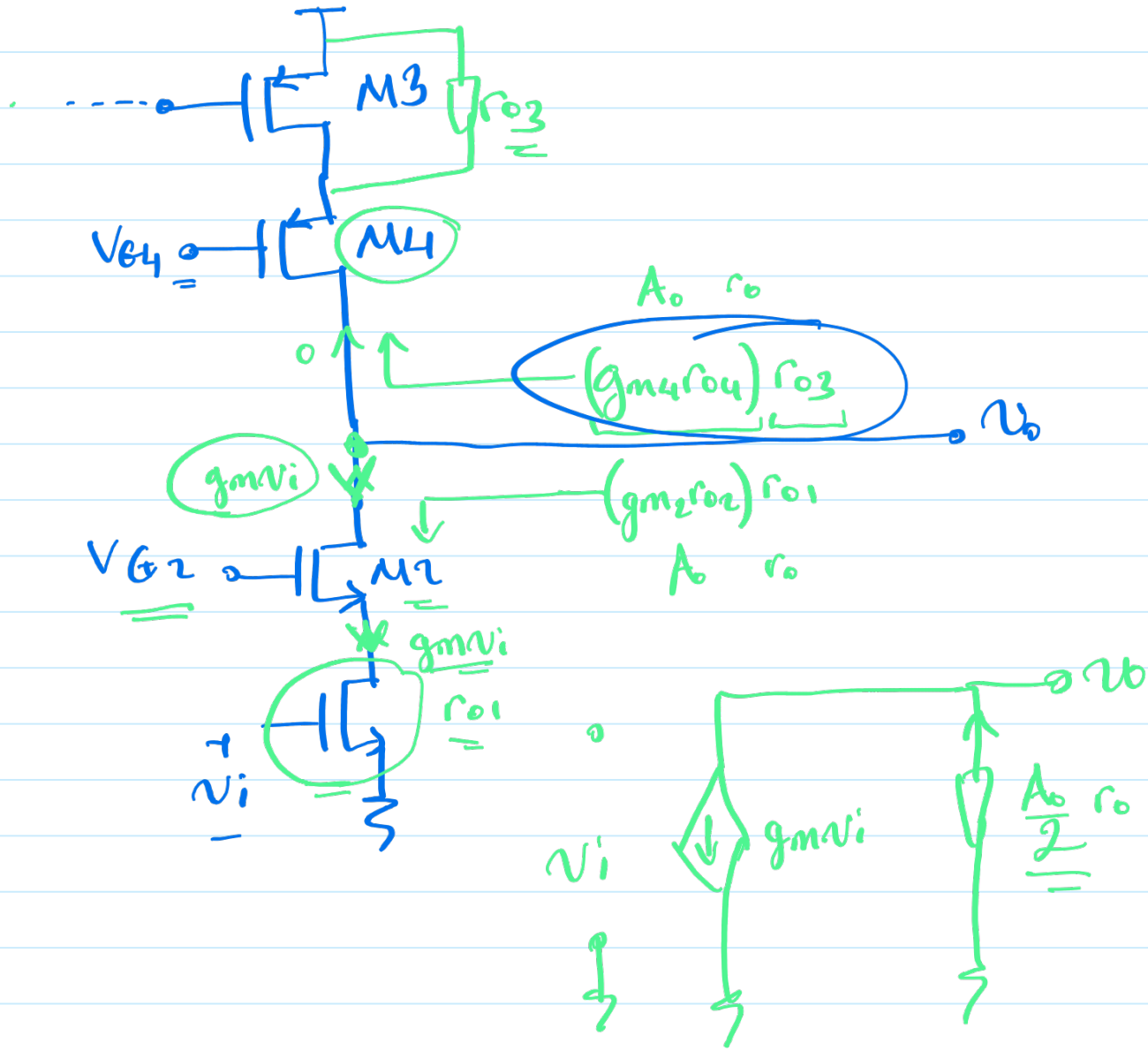




$$v_o = -g_m v_i r_{o3} \qquad \frac{v_o}{v_i} = A_v = -g_m r_{o3}$$

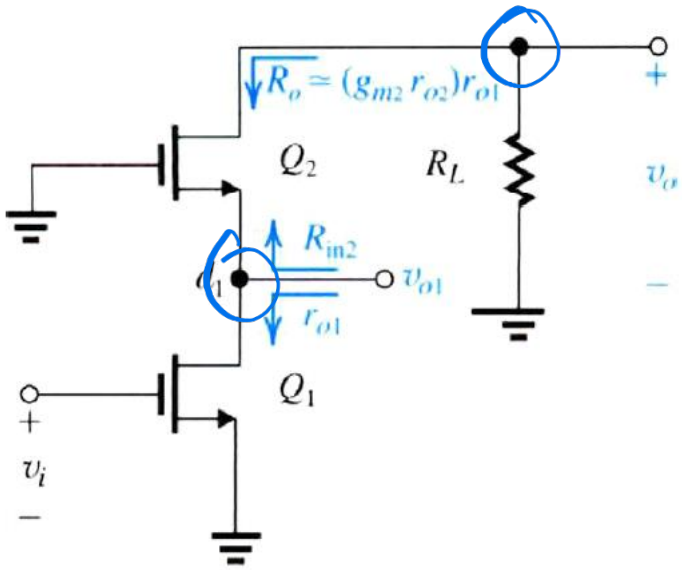
$$A_v = -A_o$$

The Cascode Amplifier with Cascode CS Load



$$v_o = -g_m v_i \frac{g_m r_o}{2} r_o \Rightarrow \frac{v_o}{v_i} = -\frac{(g_m r_o)^2}{2} = -\frac{A_0^2}{2}$$

Distribution of Voltage Gain in a Cascode Amplifier



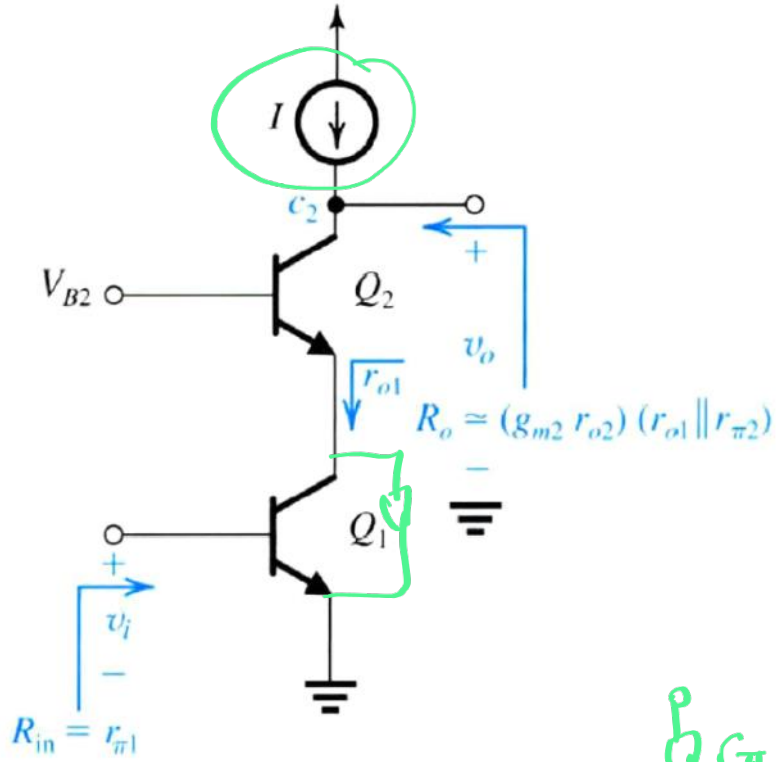
$$A_v = A_{v1} \cdot A_{v2}$$

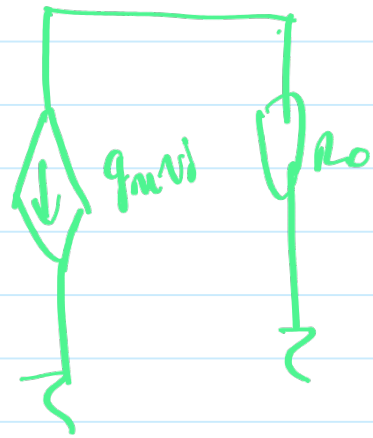
$$= \frac{v_{o1}}{v_i} \cdot \frac{v_o}{v_{o1}}$$

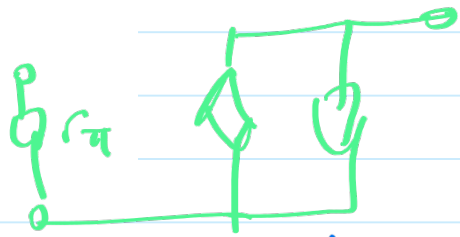
Table 8.1 Gain Distribution in the MOS Cascode Amplifier for Various Values of R_L

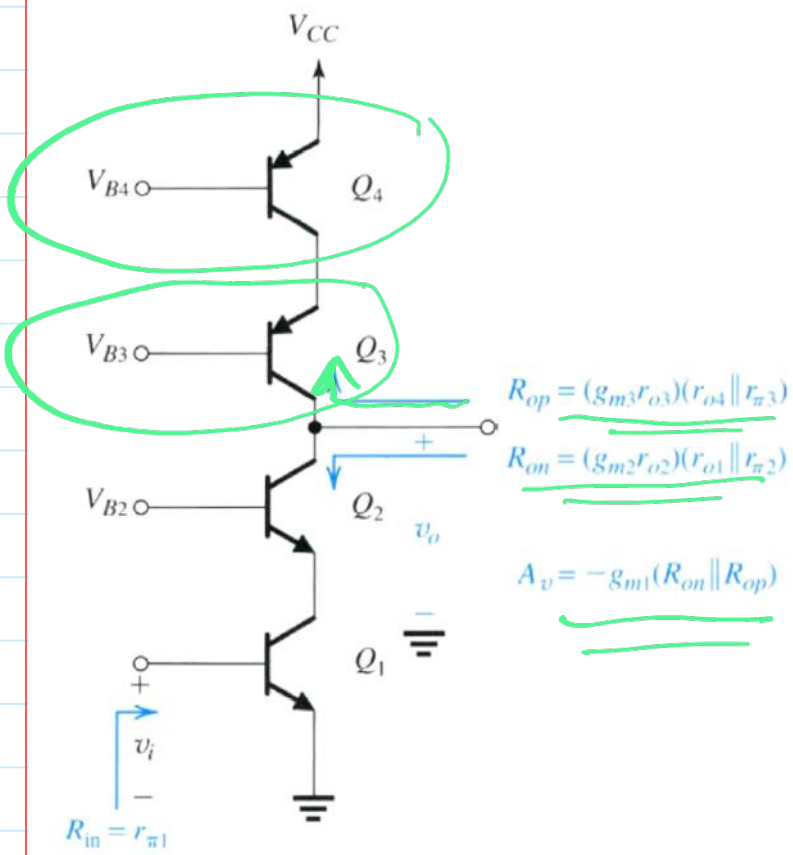
Case	R_L	R_{in2}	R_{d1}	A_{v1}	A_{v2}	A_v
1	∞	∞	r_o	$-g_m r_o$	$g_m r_o$	$-(g_m r_o)^2$
2	$(g_m r_o) r_o$	r_o	$r_o/2$	$-\frac{1}{2}(g_m r_o)$	$g_m r_o$	$-\frac{1}{2}(g_m r_o)^2$
3	r_o	$\frac{2}{g_m}$	$\frac{2}{g_m}$	-2	$\frac{1}{2}(g_m r_o)$	$-(g_m r_o)$
4	$0.2 r_0$	$\frac{1.2}{g_m}$	$\frac{1.2}{g_m}$	-1.2	$0.17(g_m r_o)$	$-0.2(g_m r_o)$

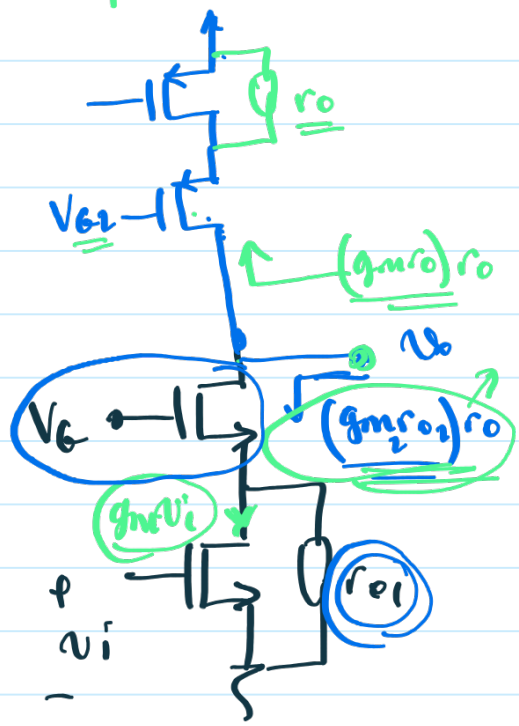
The BJT Cascode



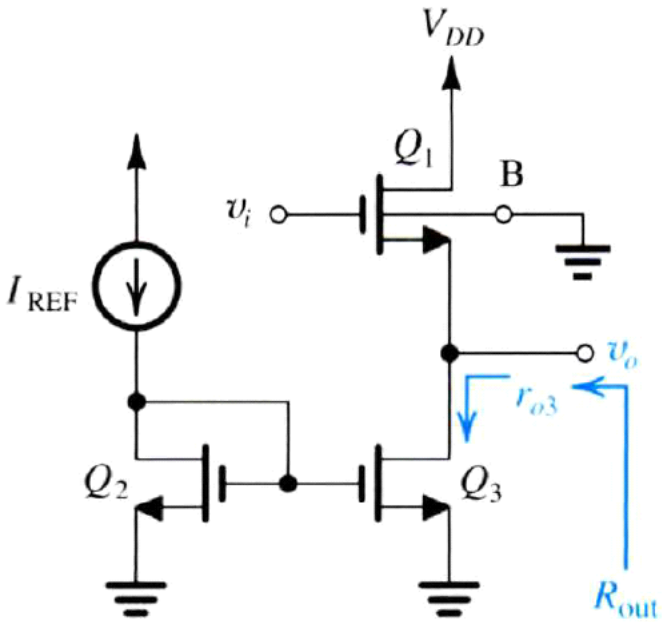


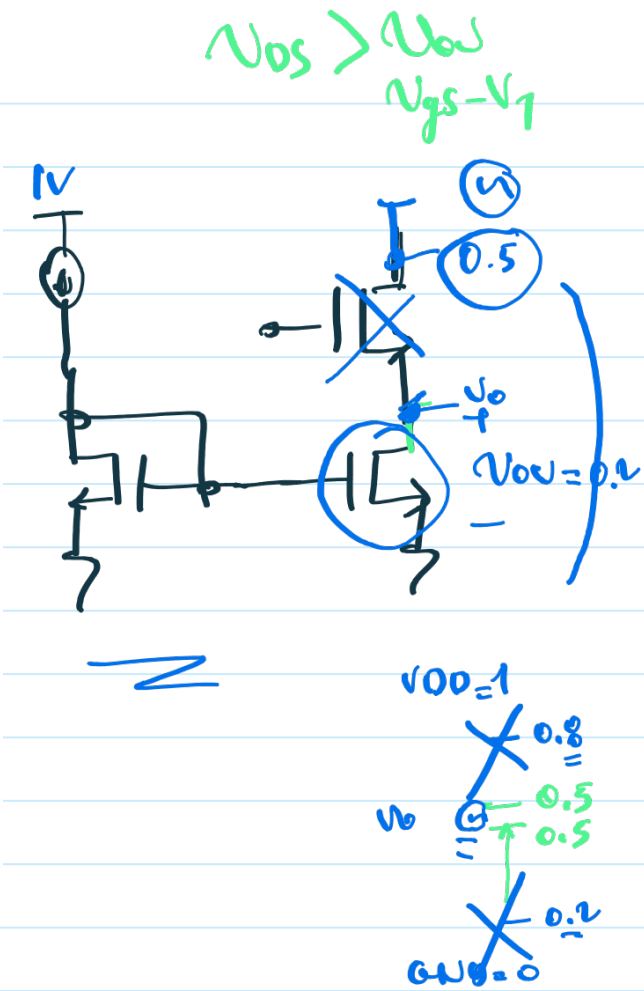




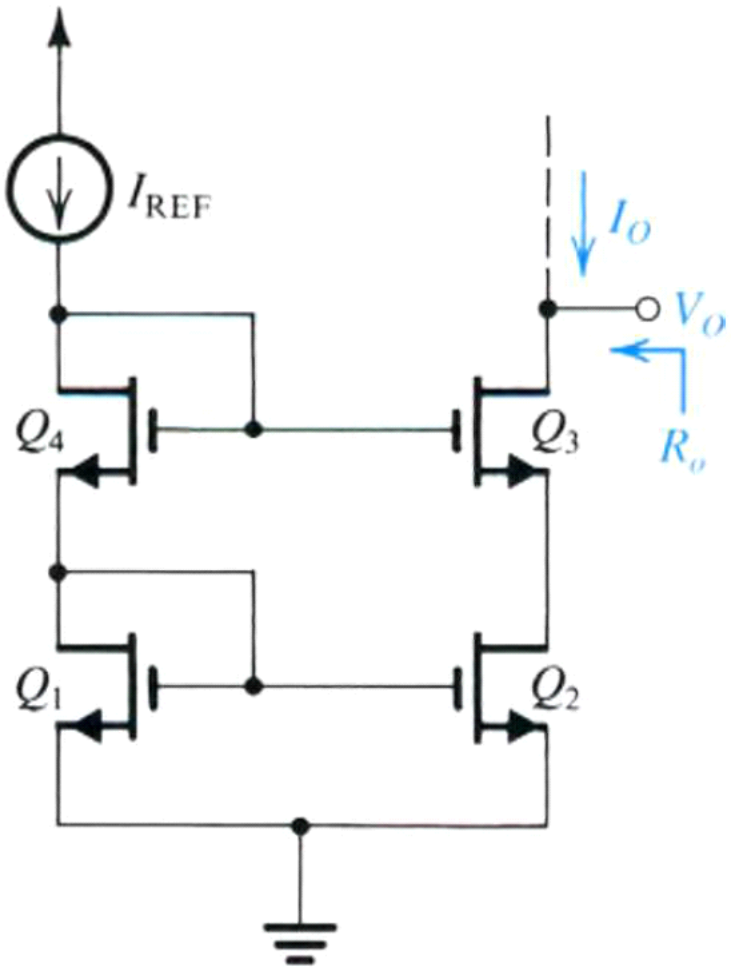


The IC Source Follower

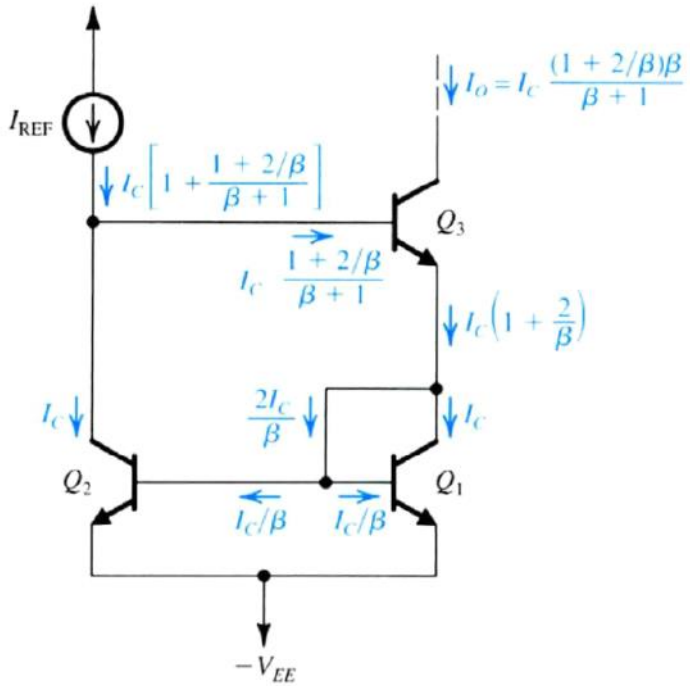




The Cascode Current Mirror

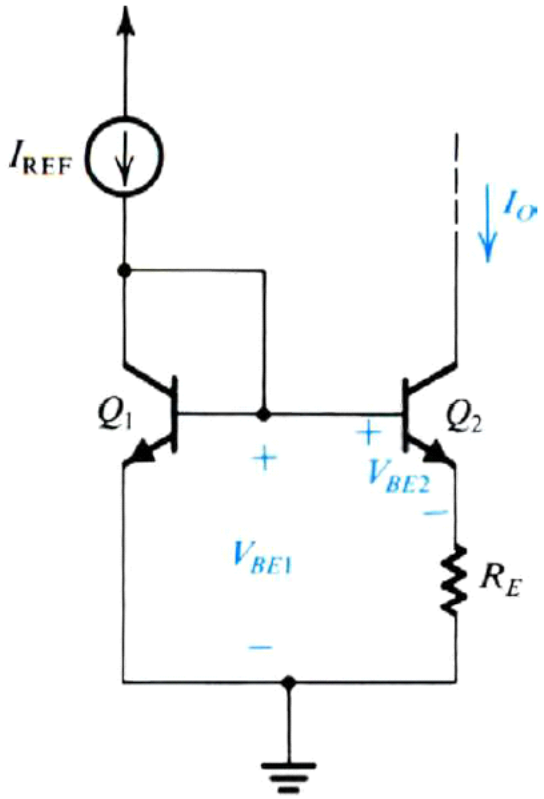


The Wilson BJT Current Mirror



The Wilson MOS Mirror

The Widlar Current Source



Template